

NICraft

*Malicious NIC Firmware-based
Cache Side-channel Attack*

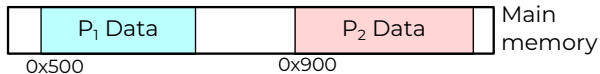
Amit Choudhari, Shorya Kumar, Christian Rossow | ESORICS 2025 | Sep 23, 2025



Background and Motivation

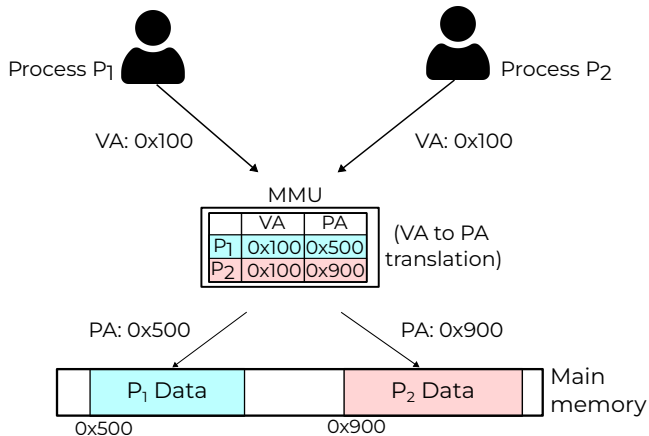


Process Isolation $\neq$ Side-Channel Isolation





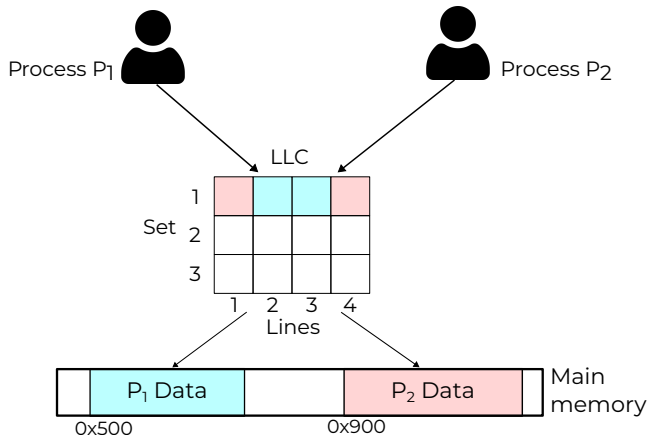
Process Isolation $\neq$ Side-Channel Isolation



Processes cannot read each other's memory (MMU / permissions)



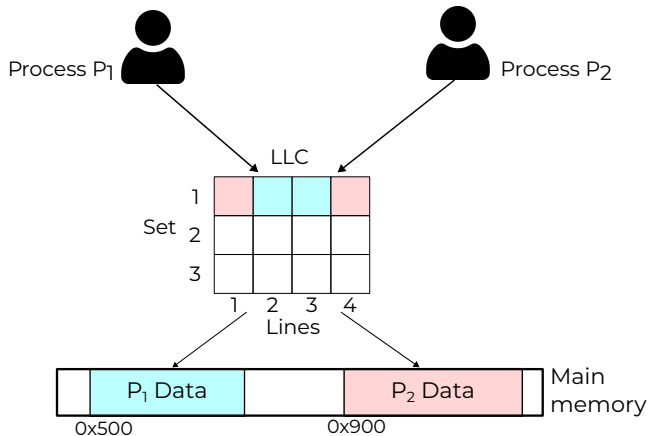
Process Isolation $\neq$ Side-Channel Isolation



But they still share the last-level cache (LLC)



Process Isolation $\neq$ Side-Channel Isolation



Risk: timing-based leakage without direct memory access



Cache Prime+Probe

Victim



Victim code

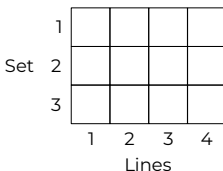
```
1  if (secret[0]) {  
2      a = set[1];  
3  } else {  
4      a = set[2];  
5  }
```

Attacker



Attacker code

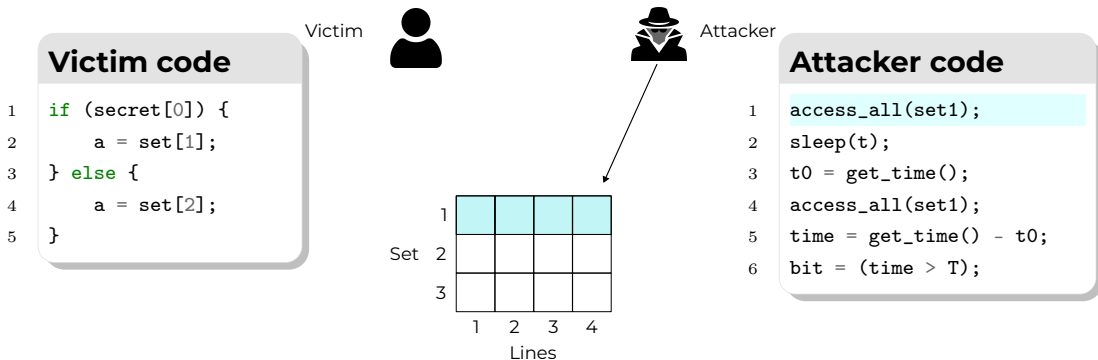
```
1  access_all(set1);  
2  sleep(t);  
3  t0 = get_time();  
4  access_all(set1);  
5  time = get_time() - t0;  
6  bit = (time > T);
```



Setup: Attacker and victim share the LLC.



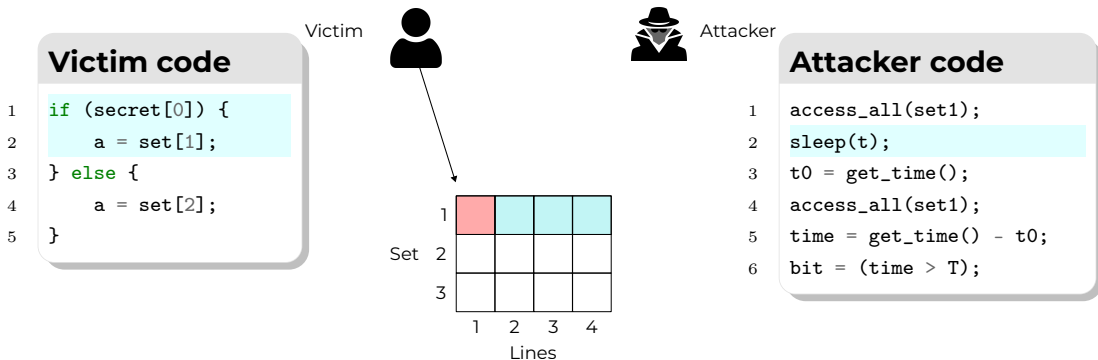
Cache Prime+Probe



Prime: Attacker access all cache lines of target set.



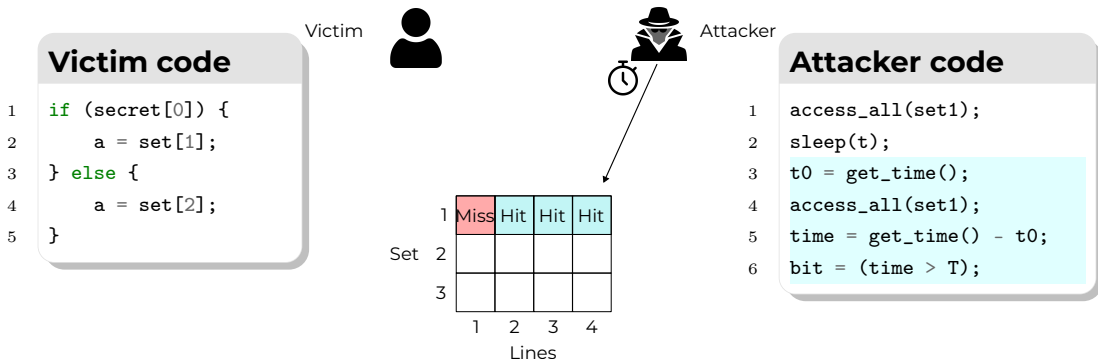
Cache Prime+Probe



Victim (secret-dependent) access: Attacker's data is evicted from the LLC.



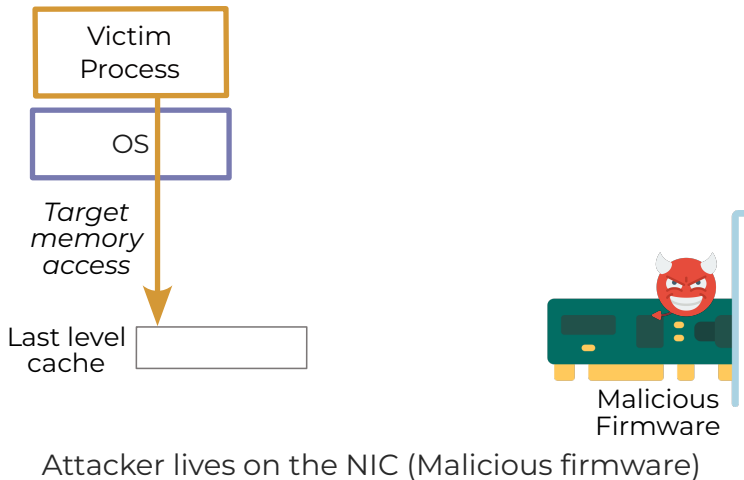
Cache Prime+Probe



Probe: Reaccess set1: **slow** (miss) $\Rightarrow$ bit=1; **fast** (hit) $\Rightarrow$ bit=0.

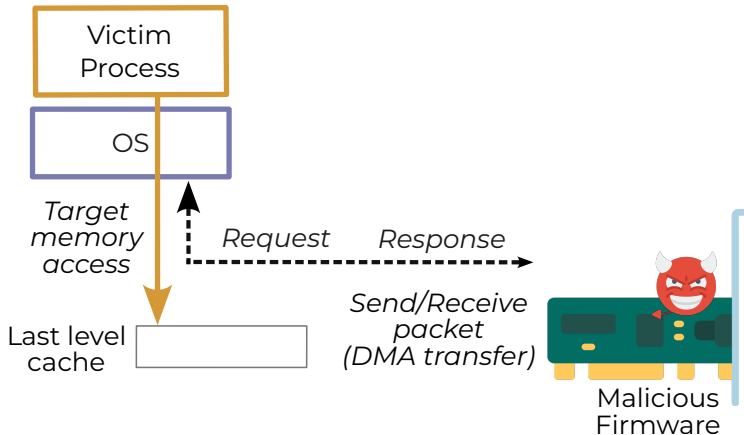


Untrusted Device: Malicious NIC Firmware





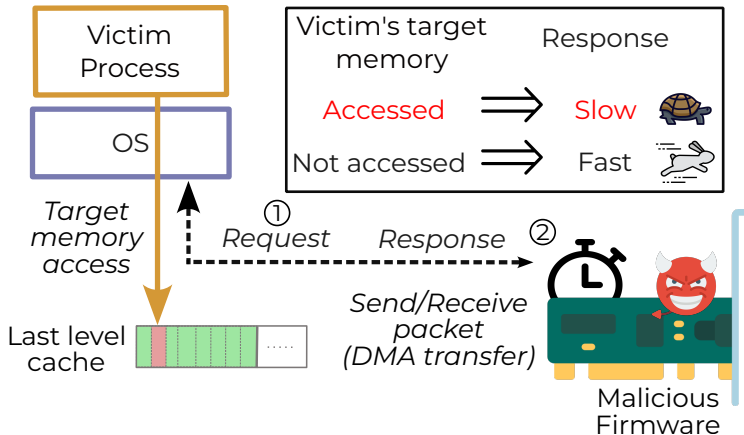
Untrusted Device: Malicious NIC Firmware



NIC can send/receive packets; craft packets at will, DMA to host memory.



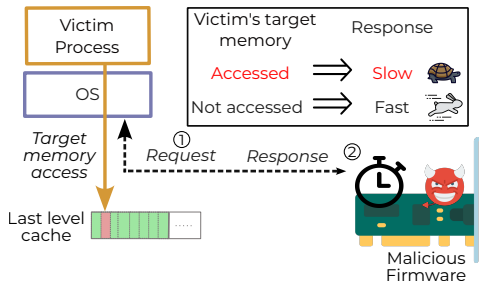
Untrusted Device: Malicious NIC Firmware



If the target memory was accessed, handling is slow; otherwise fast.



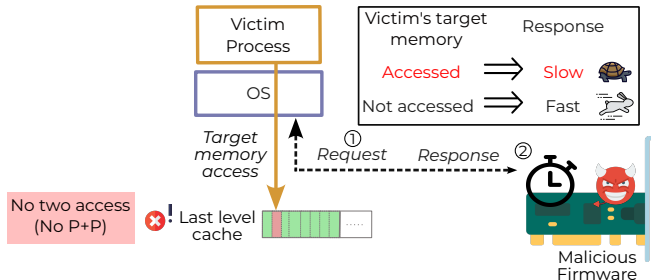
Challenges for NIC-Side Prime+Probe



Is it trivial to perform P+P cache SCA from NIC?



Challenges for NIC-Side Prime+Probe



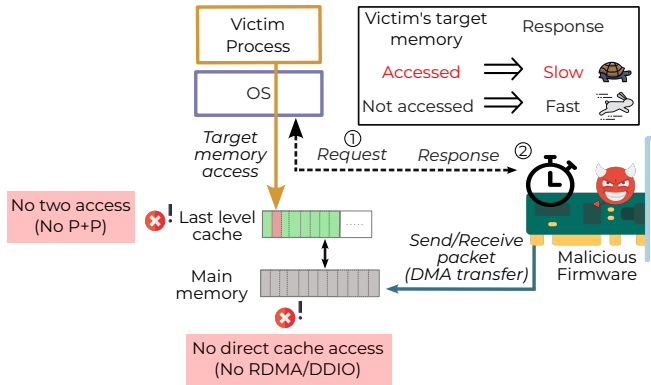
No two accesses: P + P needs two touches; we influence at most one.



P+P Mapping Coverage Control Timer



Challenges for NIC-Side Prime+Probe



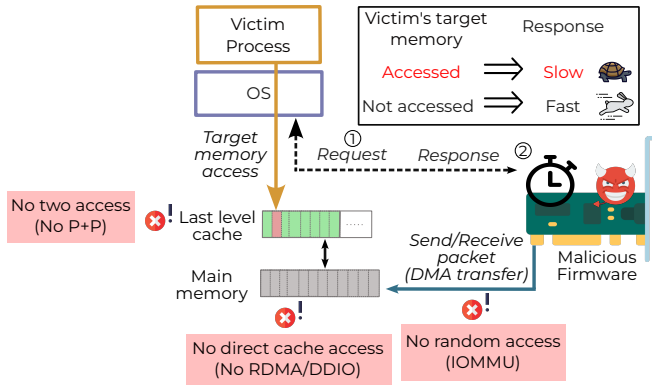
No direct cache access: NIC writes hit DRAM; kernel loads cache.



P+P Mapping Coverage Control Timer



Challenges for NIC-Side Prime+Probe



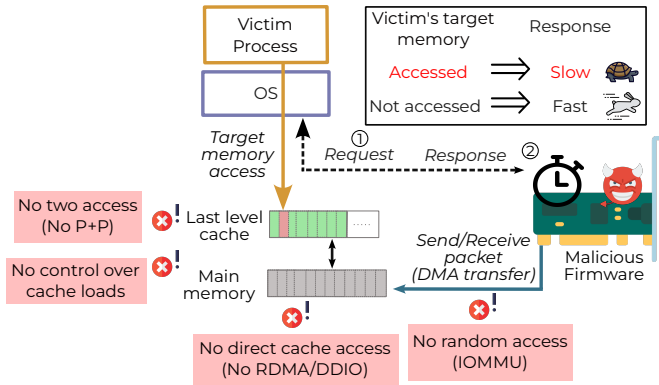
No random access: IOMMU blocks arbitrary NIC DMA.



P+P Mapping Coverage Control Timer



Challenges for NIC-Side Prime+Probe



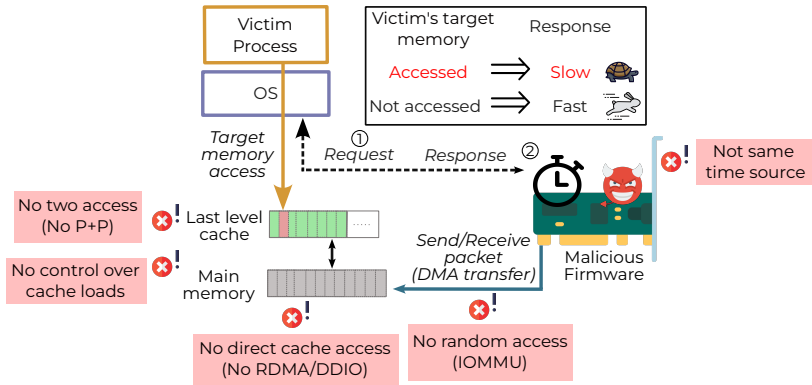
No control over loads: kernel decides when lines enter LLC.



P+P Mapping Coverage Control Timer



Challenges for NIC-Side Prime+Probe



Not same time source: NIC can't use CPU timers (`rdtsc`); need a proxy.



P+P Mapping Coverage Control Timer



NICraft Attack



Linux IP Reassembly into Prime+Probe

We exploit two kernel gadgets in IP reassembly

Prime: Header checksum

Probe: Final reassembly

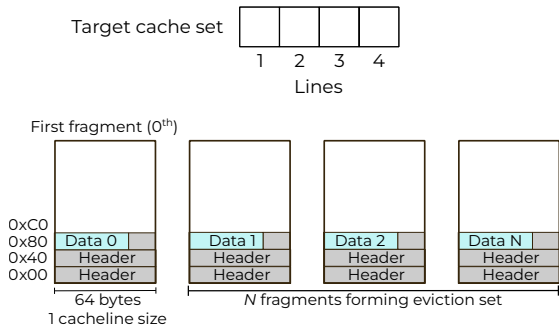


P+P Mapping Coverage Control Timer



Linux IP Reassembly into Prime+Probe

Gadget 1 (Prime): Header checksum



Header checksum

```
1 static int ip_frag_queue(struct ipq  
↪ *qp, struct sk_buff *skb) {  
2     pskb_trim_rcsum(skb, offset);  
3     if (LAST_FRAGMENT) {  
4         ip_frag_reasm(qp, skb, ...);  
5     }  
6     return 0;  
7 }
```



P+P

Mapping

Coverage

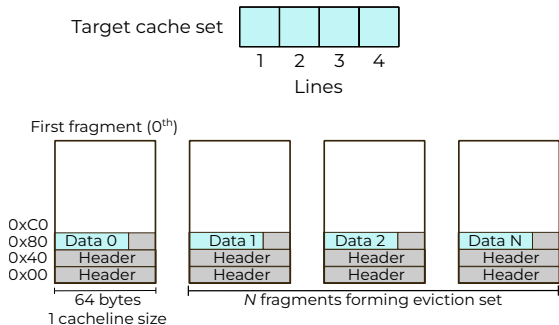
Control

Timer



Linux IP Reassembly into Prime+Probe

Gadget 1 (Prime): Header checksum



Header checksum

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P+P

Mapping

Coverage

Control

Timer



Linux IP Reassembly into Prime+Probe

Gadget 2 (Probe): Final reassembly

Fragment reassembly

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1 static int ip_frag_queue(struct ipq  
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P+P

Mapping

Coverage

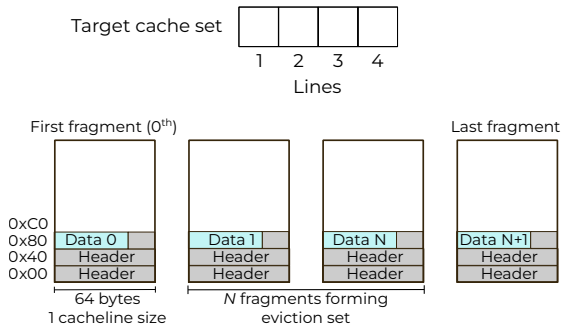
Control

Timer



Linux IP Reassembly into Prime+Probe

Gadget 2 (Probe): Final reassembly



Fragment reassembly

```
1 bool skb_try_coalesce(skb_buff *to,  
  ↪  skb_buff *from, ...)  
2 {  
3     if (len <= skb_tailroom(to)) {  
4         skb_copy_bits(from, 0,  
  ↪         skb_put(to, len), len);  
5     }  
6 }
```



P+P

Mapping

Coverage

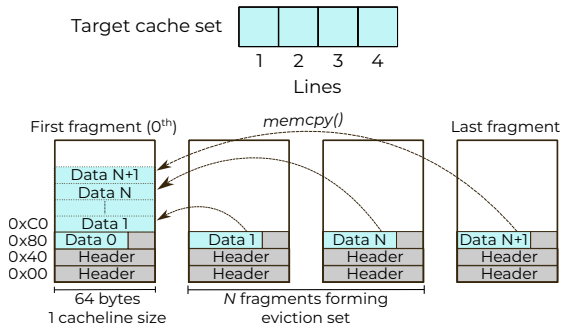
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Timer



Linux IP Reassembly into Prime+Probe

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5     }  
6 }
```



P+P

Mapping

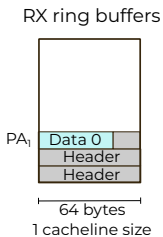
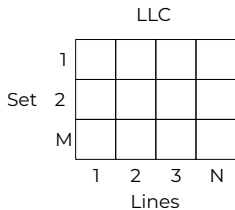
Coverage

Control

Timer



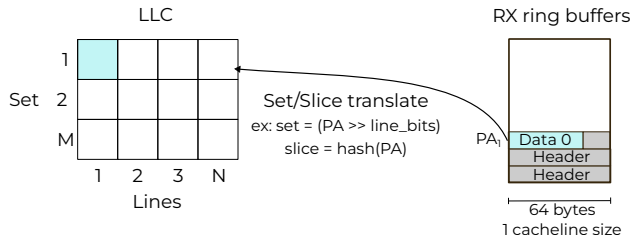
Mapping Fragments to a Target Cache Set



P+P	Mapping	Coverage	Control	Timer
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Mapping Fragments to a Target Cache Set



NIC Maps RX-buffer physical addresses → LLC set/slice.



P+P

Mapping

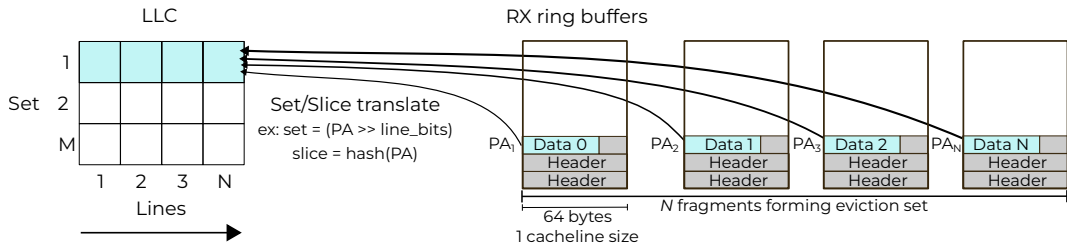
Coverage

Control

Timer



Mapping Fragments to a Target Cache Set



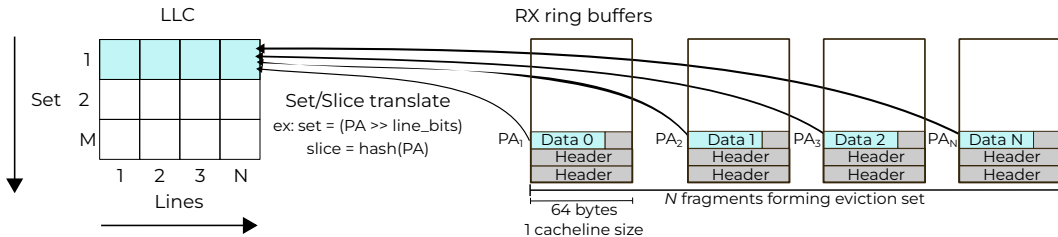
Use $N+2$ fragments on an N -way set (exclude first/last).



P+P Mapping Coverage Control Timer



Mapping Fragments to a Target Cache Set



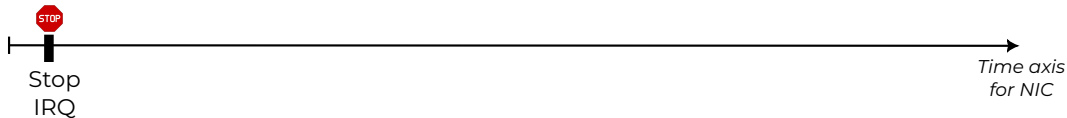
MTU/DHCP variation to expand set coverage.



P+P Mapping Coverage Control Timer



NIC Measures Victim's Access Time



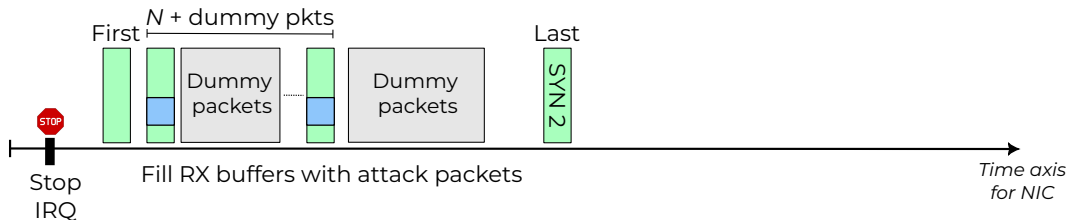
1. Mask/stop IRQs and DMA the tape without processing.



P+P Mapping Coverage Control Timer



NIC Measures Victim's Access Time



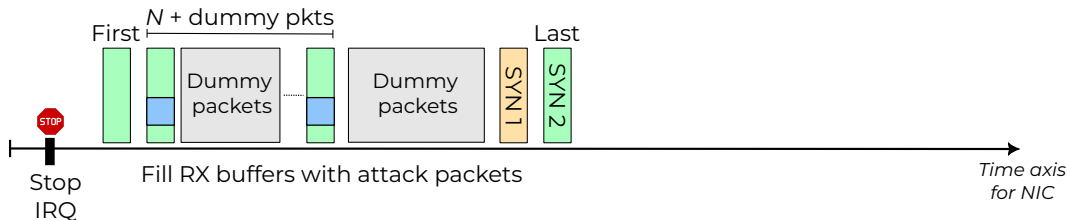
2. Fill RX ring with attack tape: SYN2 + N P-fragments + dummies.



P+P Mapping Coverage Control Timer



NIC Measures Victim's Access Time



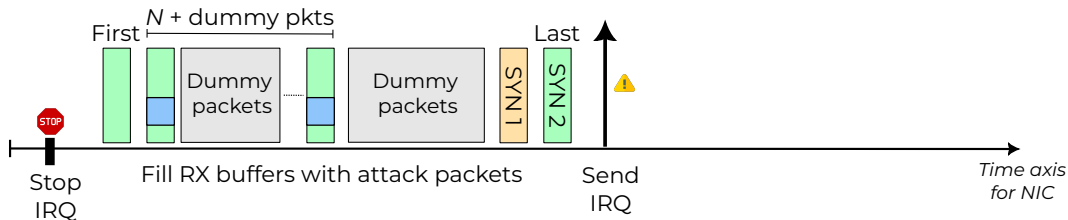
3. Insert **SYN1** for baseline timestamp t_1 .



P+P Mapping Coverage Control Timer



NIC Measures Victim's Access Time



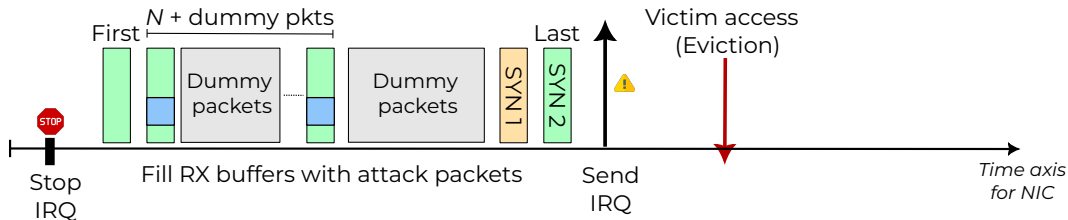
4. Raise a single IRQ; kernel processes the tape in order.



P+P Mapping Coverage Control Timer



NIC Measures Victim's Access Time



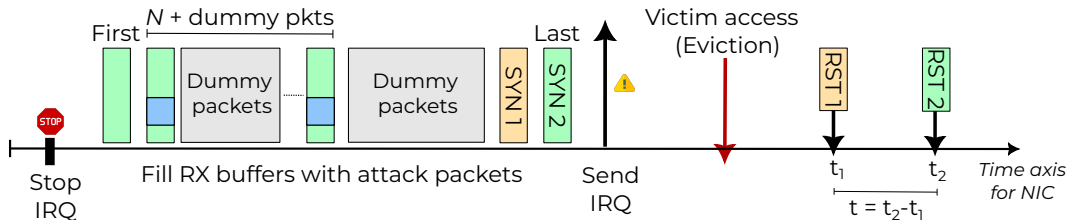
5. Victim access/eviction happens between SYN1 and SYN2.



P+P Mapping Coverage Control Timer



NIC Measures Victim's Access Time



6. Receive **RST1** then **RST2**; compute $\Delta = t_2 - t_1$ to infer eviction.



P+P Mapping Coverage Control Timer



Self-Eviction and *Aging*

Caches eviction policy (LRU): In an N -way set, inserting $> N$ distinct lines evicts an older one.



Self-Eviction and *Aging*

Self eviction

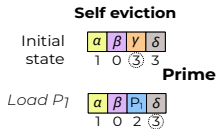
Initial	α	β	γ	δ
state	1	0	3	3

Self-eviction: Later P-lines evict earlier P-lines.

Risk: Probe becomes unreliable.



Self-Eviction and *Aging*

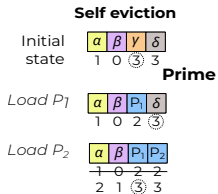


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Self-Eviction and *Aging*

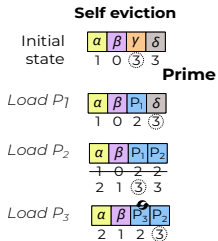


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Risk: Probe becomes unreliable.



Self-Eviction and *Aging*

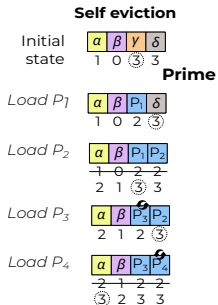


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Self-Eviction and *Aging*

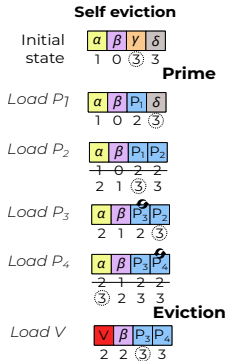


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Self-Eviction and *Aging*

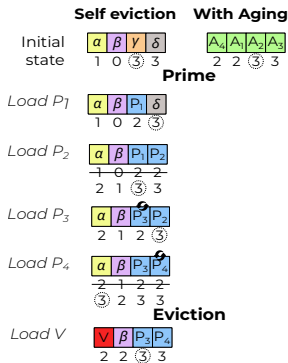


Self-eviction: Later P-lines evict earlier P-lines.

Risk: Probe becomes unreliable.



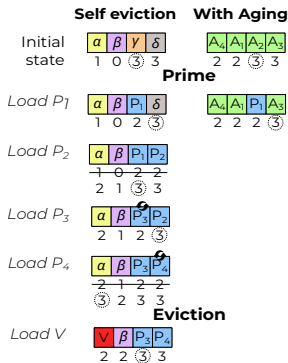
Self-Eviction and *Aging*



Aging: Preload the set with A-lines so the next N inserts evict only A's.



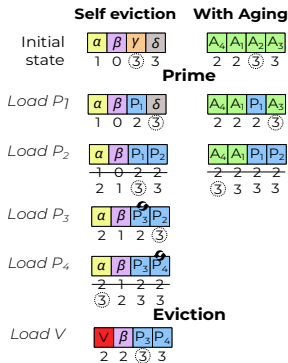
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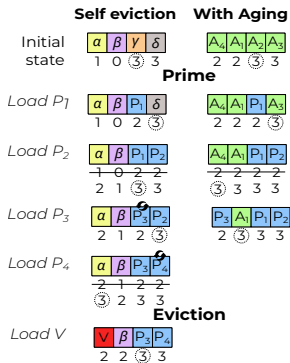
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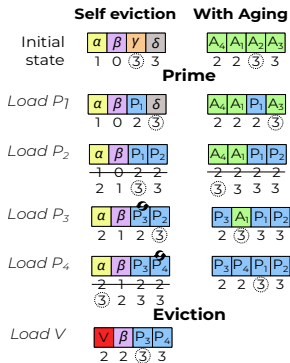
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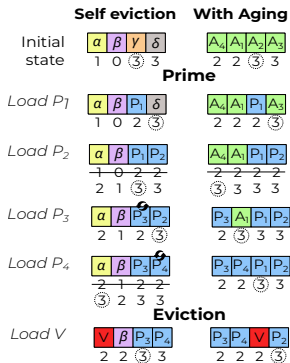
Self-Eviction and *Aging*



Aging: Preload the set with A-lines so the next N inserts evict only A's.



Self-Eviction and *Aging*



Takeaway: Aging increases the probability of correctly priming the target set.



Signal Amplification: *Domino* Effect

After **Aging**, Victim evicts exactly one P_k .
Probe sees a **single miss** on P_k .



Signal Amplification: *Domino* Effect

No Domino effect

After prime

Data	P ₄	P ₁	P ₂	P ₃
Age	3	3	3	3



Signal Amplification: *Domino* Effect

No Domino effect

After prime

Data	P ₄	P ₁	P ₂	P ₃
Age	3	3	3	3

After eviction

Load V	V	P ₁	P ₂	P ₃	Miss
	2	3	3	3	



Signal Amplification: *Domino Effect*

No Domino effect

After prime

Data	P ₄	P ₁	P ₂	P ₃
Age	3	3	3	3

After eviction

Load V	V	P ₁	P ₂	P ₃	Miss
	2	3	3	3	

Probe

Load P ₁	V	P ₁	P ₂	P ₃	Hit
	2	2	3	3	



Signal Amplification: *Domino Effect*

No Domino effect

After prime

Data	P ₄	P ₁	P ₂	P ₃
Age	3	3	3	3

After eviction

Load V	V	P ₁	P ₂	P ₃	Miss
	2	3	3	3	

Probe

Load P ₁	V	P ₁	P ₂	P ₃	Hit
	2	2	3	3	

Load P ₂	V	P ₁	P ₂	P ₃	Hit
	2	2	2	3	

Load P ₃	V	P ₁	P ₂	P ₃	Hit
	2	2	2	2	
	3	3	3	3	



Signal Amplification: *Domino* Effect

No Domino effect

After prime

Data	P ₄	P ₁	P ₂	P ₃
Age	3	3	3	3

After eviction

Load V	V	P ₁	P ₂	P ₃	Miss
	2	3	3	3	

Probe

Load P ₁	V	P ₁	P ₂	P ₃	Hit
	2	2	3	3	

Load P ₂	V	P ₁	P ₂	P ₃	Hit
	2	2	2	3	

Load P ₃	V	P ₁	P ₂	P ₃	Hit
	2	2	2	2	

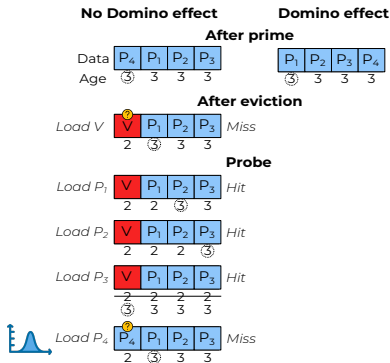
Load P ₄	P ₄	P ₁	P ₂	P ₃	Miss
	2	3	3	3	

Single miss $\Rightarrow$ small Δ signal.

We need amplification!!



Signal Amplification: *Domino* Effect

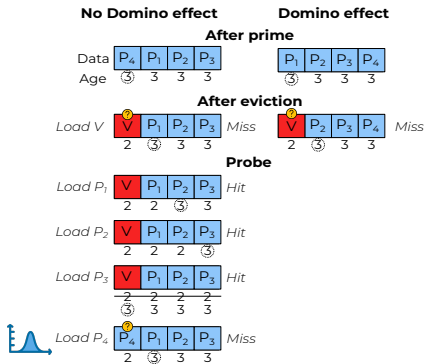


Domino: One miss triggers cascade of misses.

Order the probe accesses to mirror eviction order.

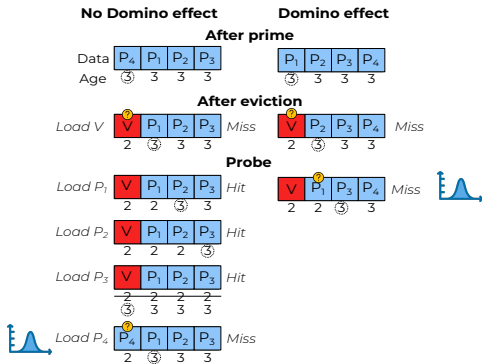


Signal Amplification: *Domino Effect*



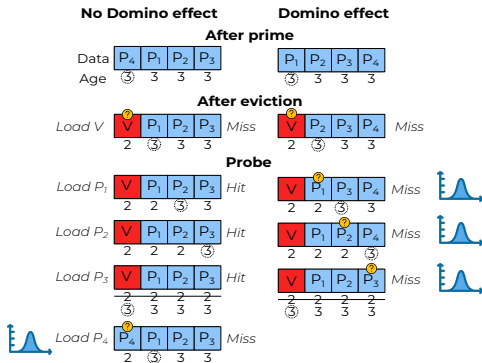


Signal Amplification: *Domino* Effect





Signal Amplification: *Domino Effect*

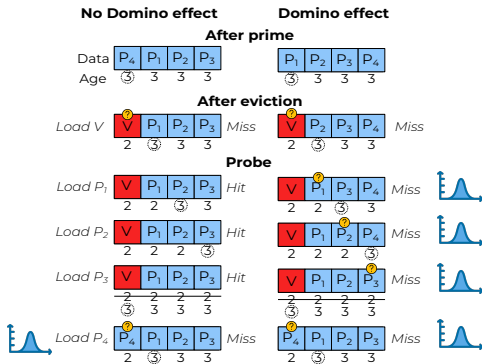


Result: larger Δ signal.

P_k miss cascades to $P_{k+1}, P_{k+2}, \dots$



Signal Amplification: *Domino* Effect



Result: larger Δ signal.

P_k miss cascades to $P_{k+1}, P_{k+2}, \dots$

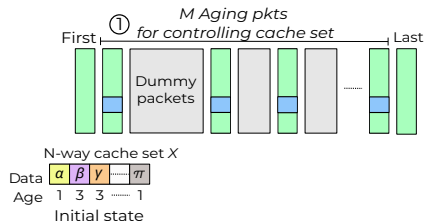


End-to-End NICraft

N-way cache set X

Data	α	β	γ		π
Age	1	3	3		1
Initial state					

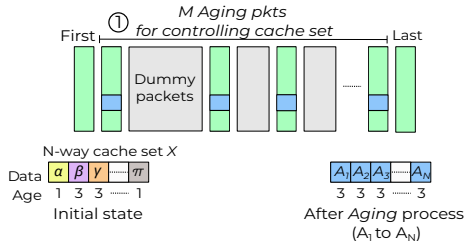
Initial cache set with random data



NIC crafts M Aging packets



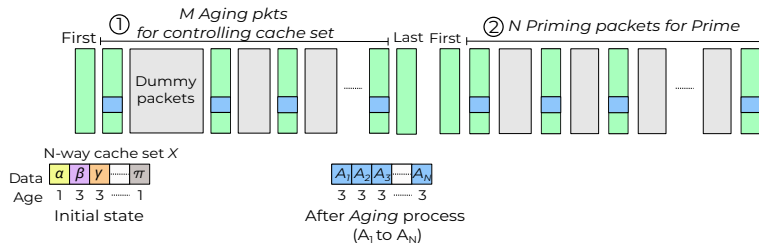
End-to-End NICraft



Aging: M packets prepares the target cache set for Priming



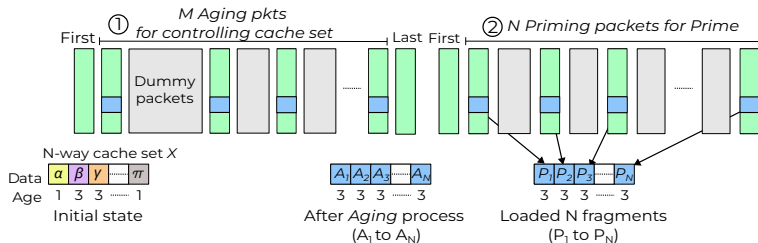
End-to-End NICraft



NIC crafts N Fragmented packets



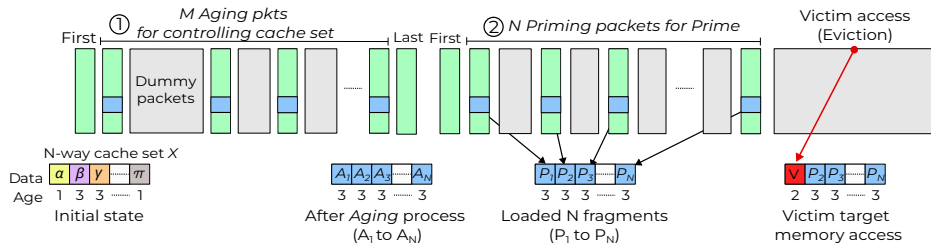
End-to-End NICraft



Prime (Gadget 1): N Fragmented packets prime the target cache set



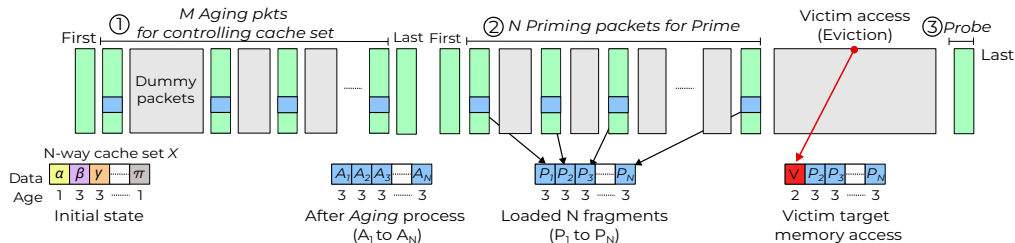
End-to-End NICraft



The victim's accesses evicts the attacker fragment data.



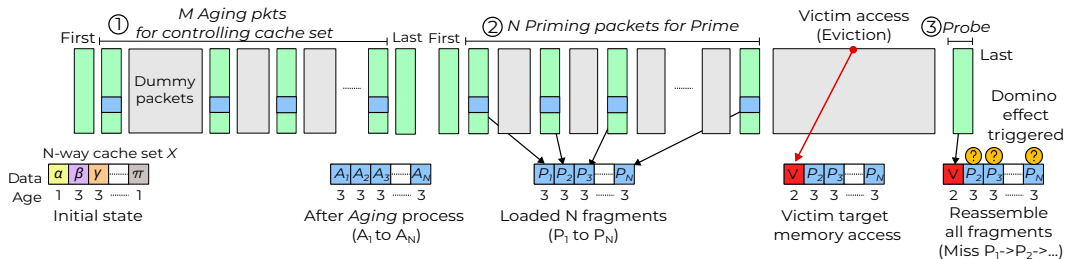
End-to-End NICraft



Probe (Gadget 2): Sending last fragmented packet triggers IP reassembly



End-to-End NICraft



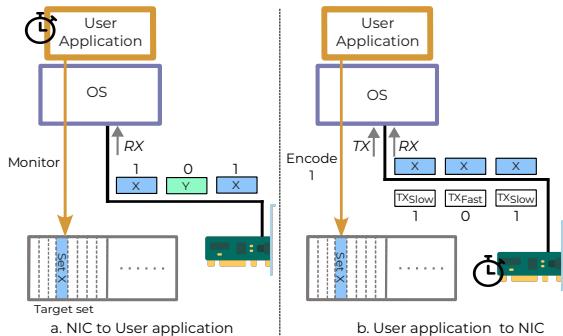
Domino effect: Domino effect kicks in due to cascading Probe misses



Case Studies

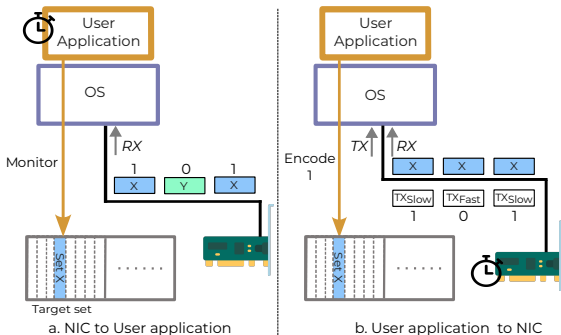


Case Study I — Covert Control Channel





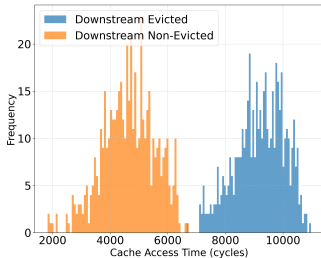
Case Study I — Covert Control Channel



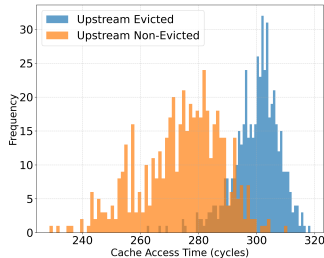
Setup: NIC firmware encodes bits by forcing/avoiding a single eviction between SYN1 and SYN2; receiver thresholds the completion-gap $\Delta = t_2 - t_1$.



Case Study I — Covert Control Channel



Downstream (NIC→Host)



Upstream (Host→NIC)

Result: Both directions **Evicted** vs **Non-evicted** are distinguishable; achieved rate ≈ 0.76 bit/s (NIC→Host) and ≈ 0.10 bit/s (Host→NIC).



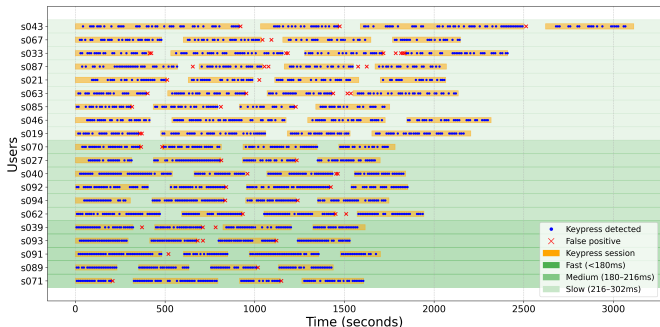
Case Study II — Keystroke Activity Detection

Goal: Detect **typing activity** (not content) from the NIC.

Result: We detect typing activity from the NIC's with $\geq 96\%$ **precision** overall.



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Mitigations and Conclusion



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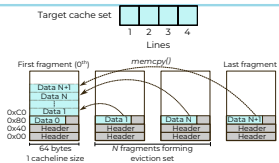
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Disrupt predictability

- Randomize network buffer placement
- **Flush header cache line (fragmented pkts)**



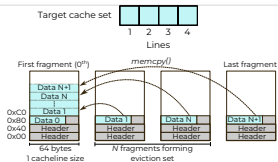
Summary



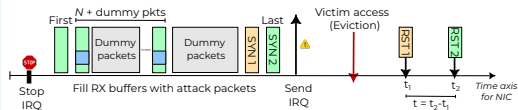
Prime+Probe from Reassembly



Summary



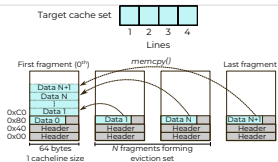
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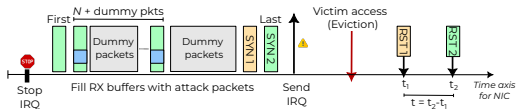
One-IRQ Mechanism (Timing)



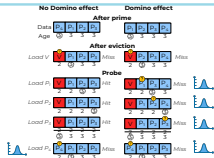
Summary



Prime+Probe from Reassembly



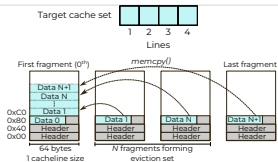
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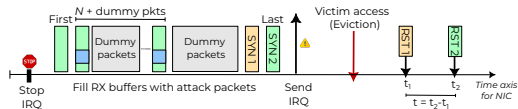
Aging + Signal Amplification (Domino)



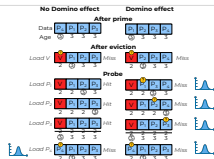
Summary



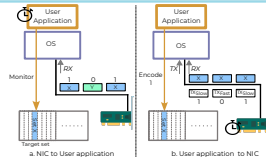
Prime+Probe from Reassembly



One-IRQ Mechanism (Timing)



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Case Studies